main.cpp

```
        if (!(cin >> choice)) {
            cout << "Please enter a number.\n";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
            continue;
        }

        cin.ignore(numeric_limits<streamsize>::max(), '\n');

        switch (choice) {
            case 1: bookTicket(); break;
            case 2: cancelTicket(); break;
            case 3: viewSeatStatus(); break;
            case 4: passengerList(); break;
            case 5: searchPassenger(); break;
            case 6: simulateDisruption(); break;
            case 7: recoveryReport(); break;
            case 8:
                cleanup();
                cout << "RaahSathi closed. Memory released.\n";
                break;
            default:
                cout << "Invalid choice. Select 1-8.\n";
        }
    } while (choice != 8);

    return 0;
}
```

Input

```
2. Cancel Ticket
3. View Seat Status
4. View Passenger List
5. Search Passenger
6. Simulate Railway Disruption
7. Passenger Recovery Report
8. Exit
Enter choice:
```

main.cpp

```cpp
#include <iostream>
#include <string>
#include <cstdlib>
#include <new>
#include <limits>
using namespace std;

const int TOTAL_SEATS = 30;

// Seat: 0 = Available, 1 = Booked, 2 = Blocked
int* seats = nullptr;

struct Passenger {
    string name;
    int age;
    int priority; // 1 = Urgent, 2 = Normal
    int seat;     // 0 = Waiting List
    Passenger* next;
};

Passenger* head = nullptr;

Passenger* createPassenger(string name, int age,
                           int priority, int seat) {
    void* memory = malloc(sizeof(Passenger));

    if (memory == nullptr)
```

main.cpp

```
28          return nullptr;
29
30      Passenger* p = new(memory) Passenger;
31      p->name = name;
32      p->age = age;
33      p->priority = priority;
34      p->seat = seat;
35      p->next = nullptr;
36
37      return p;
38  }
39
40  void addPassenger(Passenger* p) {
41      if (head == nullptr) {
42          head = p;
43          return;
44      }
45
46      Passenger* temp = head;
47      while (temp->next != nullptr)
48          temp = temp->next;
49
50      temp->next = p;
51  }
52
53  int findAvailableSeat() {
54      for (int i = 0; i < TOTAL_SEATS; i++)
```

```
55          if (seats[i] == 0)
56              return i + 1;
57
58      return 0;
59  }
60
61  void promoteWaitingPassengers() {
62      while (true) {
63          int available = findAvailableSeat();
64          if (available == 0)
65              return;
66
67          Passenger* best = nullptr;
68
69          for (Passenger* p = head; p != nullptr; p = p->next) {
70              if (p->seat == 0 &&
71                  (best == nullptr || p->priority < best->priority))
72                  best = p;
73          }
74
75          if (best == nullptr)
76              return;
77
78          best->seat = available;
79          seats[available - 1] = 1;
80
81          cout << "Waitlisted passenger " << best->name
```

main.cpp

```
            best->seat = available;
            seats[available - 1] = 1;

            cout << "Waitlisted passenger " << best->name
                 << " allocated seat " << available << ".\n";
        }
    }

void bookTicket() {
    string name;
    int age, priority;

    cout << "Passenger name: ";
    getline(cin, name);

    cout << "Age: ";
    if (!(cin >> age) || age < 1 || age > 120) {
        cout << "Invalid age.\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
        return;
    }

    cout << "Priority (1=Urgent, 2=Normal): ";
    if (!(cin >> priority) || (priority != 1 && priority != 2)) {
        cout << "Invalid priority.\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
```

input

main.cpp

```
98          return;
99      }
100
101     cout << "Priority (1=Urgent, 2=Normal): ";
102     if (!(cin >> priority) || (priority != 1 && priority != 2)) {
103         cout << "Invalid priority.\n";
104         cin.clear();
105         cin.ignore(numeric_limits<streamsize>::max(), '\n');
106         return;
107     }
108     cin.ignore(numeric_limits<streamsize>::max(), '\n');
109
110     int seat = findAvailableSeat();
111     Passenger* p = createPassenger(name, age, priority, seat);
112
113     if (p == nullptr) {
114         cout << "Memory allocation failed.\n";
115         return;
116     }
117
118     addPassenger(p);
119
120     if (seat != 0) {
121         seats[seat - 1] = 1;
122         cout << "Ticket confirmed! Seat: " << seat << "\n";
123     } else {
124         cout << "All seats full. Added to waiting list.\n";
125     }
```

input

main.cpp

```
125     }
126 }
127
128 void cancelTicket() {
129     int seat;
130     cout << "Enter booked seat number: ";
131
132     if (!(cin >> seat)) {
133         cout << "Invalid input.\n";
134         cin.clear();
135         cin.ignore(numeric_limits<streamsize>::max(), '\n');
136         return;
137     }
138     cin.ignore(numeric_limits<streamsize>::max(), '\n');
139
140     if (seat < 1 || seat > TOTAL_SEATS || seats[seat - 1] != 1) {
141         cout << "That seat is not booked.\n";
142         return;
143     }
144
145     Passenger* prev = nullptr;
146     Passenger* p = head;
147
148     while (p != nullptr && p->seat != seat) {
149         prev = p;
150         p = p->next;
151     }
```

```
152
153     if (p == nullptr) {
154         cout << "Passenger record not found.\n";
155         return;
156     }
157
158     seats[seat - 1] = 0;
159
160     if (prev == nullptr)
161         head = p->next;
162     else
163         prev->next = p->next;
164
165     cout << "Ticket cancelled for " << p->name << ".\n";
166
167     p->Passenger();
168     free(p);
169
170     promoteWaitingPassengers();
171 }
172
173 void viewSeatStatus() {
174     cout << "\n--- Seat Map ---\n";
175
176     for (int i = 0; i < MAX_SEATS; i++) {
177         cout << "Seat " << i + 1 << ": ";
178
```

```
178
179         if (seats[i] == 0) cout << "Available";
180         else if (seats[i] == 1) cout << "Booked";
181         else cout << "Blocked";
182
183         cout << ((i + 1) % 3 == 0 ? "\n" : " | ");
184     }
185 }
186
187 void passengerList() {
188     cout << "\n--- PASSENGER LIST ---\n";
189
190     if (head == nullptr) {
191         cout << "No passenger records.\n";
192         return;
193     }
194
195     for (Passenger* p = head; p != nullptr; p = p->next) {
196         cout << "Name: " << p->name
197              << " | Age: " << p->age
198              << " | Priority: " << p->priority
199              << " | ";
200
201         if (p->seat == 0)
202             cout << "Waiting List";
203         else
204             cout << "Seat: " << p->seat;
```

main.cpp

```
205
206         cout << "\n";
207     }
208 }
209
210 void searchPassenger() {
211     string name;
212     cout << "Enter passenger name to search: ";
213     getline(cin, name);
214
215     bool found = false;
216
217     for (Passenger* p = head; p != nullptr; p = p->next) {
218         if (p->name == name) {
219             cout << "Found: " << p->name
220                  << " | Age: " << p->age
221                  << " | Priority: " << p->priority
222                  << " | Seat: ";
223
224             if (p->seat == 0) cout << "Waiting List\n";
225             else cout << p->seat << "\n";
226
227             found = true;
228         }
229     }
230
231     if (!found)
232         cout << "Passenger not found.\n";
```

```
230
231     if (!found)
232         cout << "Passenger not found.\n";
233 }
234
235 void simulateDisruption() {
236     int seat;
237     cout << "Enter seat number affected by disruption: ";
238
239     if (!(cin >> seat)) {
240         cout << "Invalid input.\n";
241         cin.clear();
242         cin.ignore(numeric_limits<streamsize>::max(), '\n');
243         return;
244     }
245     cin.ignore(numeric_limits<streamsize>::max(), '\n');
246
247     if (seat < 1 || seat > TOTAL_SEATS || seats[seat - 1] == 2) {
248         cout << "Invalid seat or seat already blocked.\n";
249         return;
250     }
251
252     Passenger* affected = nullptr;
253
254     if (seats[seat - 1] == 1) {
255         for (Passenger* p = head; p != nullptr; p = p->next) {
256             if (p->seat == seat) {
```

++_compiler

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main.cpp

```
257                 affected = p;
258                 break;
259             }
260         }
261
262         if (affected == nullptr) {
263             cout << "Passenger record not found.\n";
264             return;
265         }
266     }
267
268     seats[seat - 1] = 2;
269     cout << "Disruption simulated: Seat " << seat << " blocked.\n";
270
271     if (affected != nullptr) {
272         int alternative = findAvailableSeat();
273
274         if (alternative != 0) {
275             affected->seat = alternative;
276             seats[alternative - 1] = 1;
277
278             cout << affected->name << " moved from seat "
279                  << seat << " to seat " << alternative << ".\n";
280         } else {
281             affected->seat = 0;
282             cout << affected->name
283                  << " added to waiting list: no alternative seat.\n";
```

main.cpp

```
                << seat << " to seat " << alternative << ".\n";
        } else {
            affected->seat = 0;
            cout << affected->name
                 << " added to waiting list: no alternative seat.\n";
        }
    } else {
        cout << "No passenger was affected by this seat disruption.\n";
    }
}

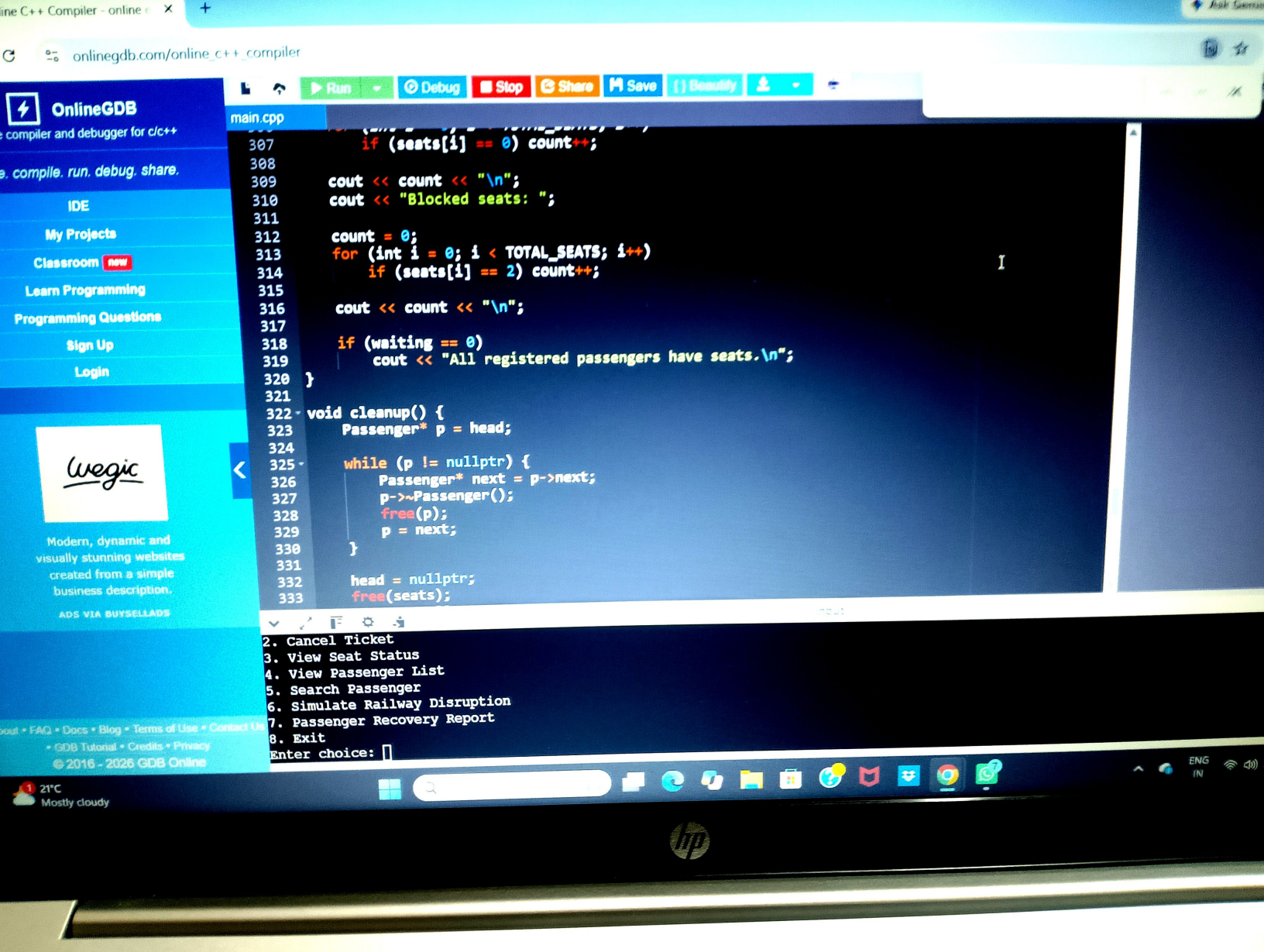
void recoveryReport() {
    cout << "\n--- RECOVERY REPORT ---\n";
    int waiting = 0;

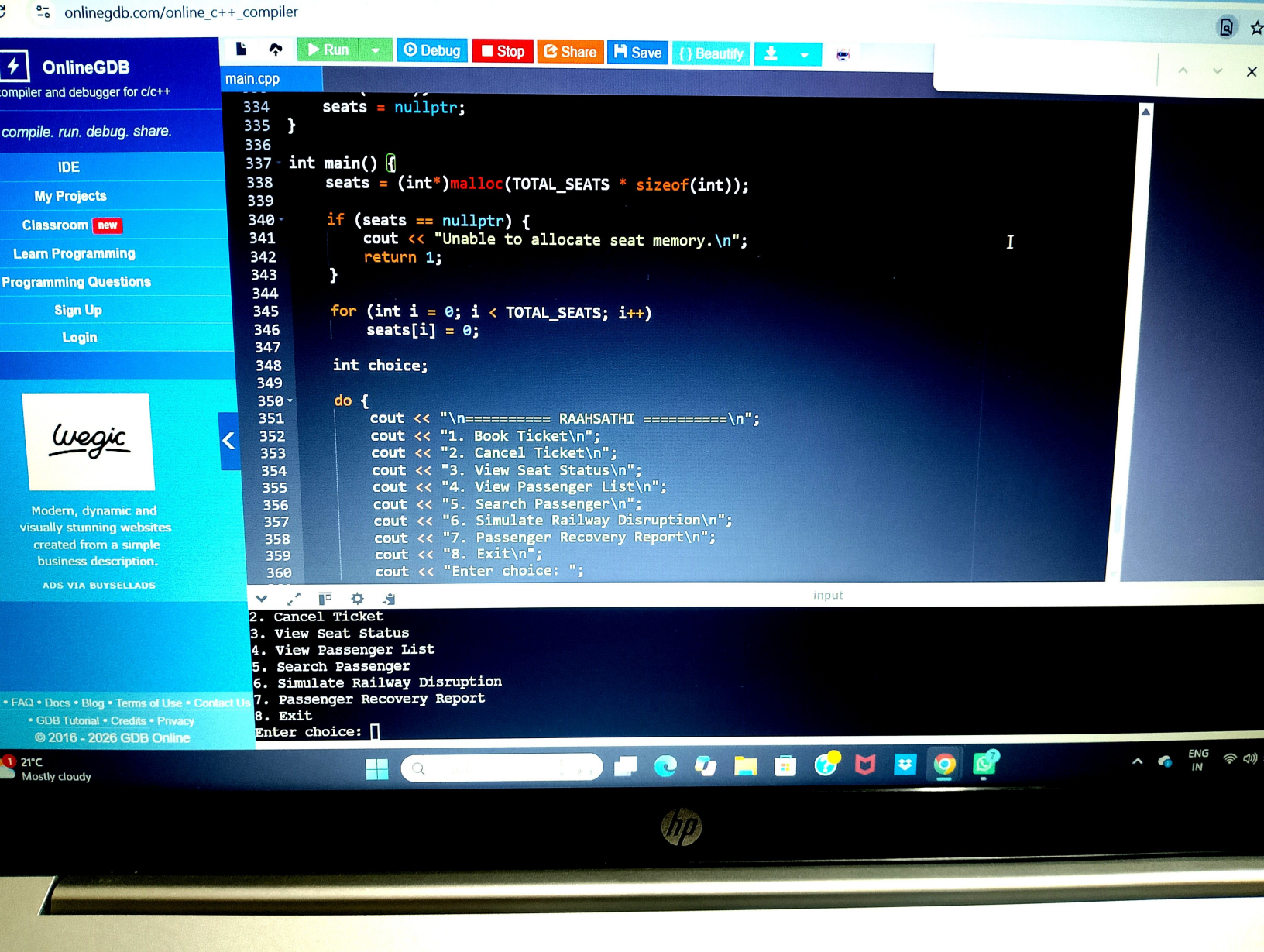
    for (Passenger* p = head; p != nullptr; p = p->next) {
        if (p->seat == 0) {
            cout << "WAITING: " << p->name
                 << " | Priority: " << p->priority << "\n";
            waiting++;
        }
    }

    cout << "Passengers waiting: " << waiting << "\n";
    cout << "Available seats: ";

    int count = 0;
    for (int i = 0; i < TOTAL_SEATS; i++)
```

input


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main.cpp
if (seats[i] == 0) count++;
cout << count << "\n";
cout << "Blocked seats: ";
count = 0;
for (int i = 0; i < TOTAL_SEATS; i++)
if (seats[i] == 2) count++;
cout << count << "\n";
if (waiting == 0)
cout << "All registered passengers have seats.\n";
}
void cleanup() {
Passenger* p = head;
while (p != nullptr) {
Passenger* next = p->next;
p->~Passenger();
free(p);
p = next;
}
head = nullptr;
free(seats);
2. Cancel Ticket
3. View Seat Status
4. View Passenger List
5. Search Passenger
6. Simulate Railway Disruption
7. Passenger Recovery Report
8. Exit
Enter choice:
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ENG
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{ } Beautify
main.cpp
seats = nullptr;
}
int main() {
seats = (int*)malloc(TOTAL_SEATS * sizeof(int));
if (seats == nullptr) {
cout << "Unable to allocate seat memory.\n";
return 1;
}
for (int i = 0; i < TOTAL_SEATS; i++)
seats[i] = 0;
int choice;
do {
cout << "\n========== RAAHSATHI ==========\n";
cout << "1. Book Ticket\n";
cout << "2. Cancel Ticket\n";
cout << "3. View Seat Status\n";
cout << "4. View Passenger List\n";
cout << "5. Search Passenger\n";
cout << "6. Simulate Railway Disruption\n";
cout << "7. Passenger Recovery Report\n";
cout << "8. Exit\n";
cout << "Enter choice: ";
input
2. Cancel Ticket
3. View Seat Status
4. View Passenger List
5. Search Passenger
6. Simulate Railway Disruption
7. Passenger Recovery Report
8. Exit
Enter choice:
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IN

```
362         if (!(cin >> choice)) {
363             cout << "Please enter a number.\n";
364             cin.clear();
365             cin.ignore(numeric_limits<streamsize>::max(), '\n');
366             continue;
367         }
368
369         cin.ignore(numeric_limits<streamsize>::max(), '\n');
370
371         switch (choice) {
372             case 1: bookTicket(); break;
373             case 2: cancelTicket(); break;
374             case 3: viewSeatStatus(); break;
375             case 4: passengerList(); break;
376             case 5: searchPassenger(); break;
377             case 6: simulateDisruption(); break;
378             case 7: recoveryReport(); break;
379             case 8:
380                 cleanup();
381                 cout << "RaahSathi closed. Memory released.\n";
382                 break;
383             default:
384                 cout << "Invalid choice. Select 1-8.\n";
385         }
386     } while (choice != 8);
387
388     return 0;
```

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```
365             cin.ignore(numeric_limits<streamsize>::max(), '\n');
366             continue;
367         }
368
369         cin.ignore(numeric_limits<streamsize>::max(), '\n');
370
371         switch (choice) {
372             case 1: bookTicket(); break;
373             case 2: cancelTicket(); break;
374             case 3: viewSeatStatus(); break;
375             case 4: passengerList(); break;
376             case 5: searchPassenger(); break;
377             case 6: simulateDisruption(); break;
378             case 7: recoveryReport(); break;
379             case 8:
380                 cleanup();
381                 cout << "RaahSathi closed. Memory released.\n";
382                 break;
383             default:
384                 cout << "Invalid choice. Select 1-8.\n";
385         }
386     } while (choice != 8);
387
388     return 0;
389 }
```